

ISYE 7406 HW1

Introduction

Problem Description

This assignment compares the classification methods for the handwritten numbers “2” and “7.” Specifically, it looks at linear regression and k-nearest neighbors (KNN) and their effectiveness in classifying these two numbers, evaluating which is better.

A label, Y (denoted as $V1$ in R), indicates whether the number is a “2” or “7.” Anything other than these two in the category is excluded from the dataset. There are also 256 ($V2$ to $V257$) independent variables, which serve as the pixels of a 16x16-dimensional illustration of the numbers.

Data Set

```
> dim(ziptrain27); ## 1376 257
[1] 1376 257
ziptest27 <- subset(ziptest, ziptest[,1]==2 | ziptest[,1]==7);
dim(ziptest27) ##345 257
```

After filtering out the dataset to only include $Y = 2$ and $Y = 7$, the training data consists of 1,376 examples, while the testing data contains 345 examples, designed with the 75/25 split in mind. Monte Carlo Cross-validation is also used to ensure that the results are not due to a single chance of a split in data.

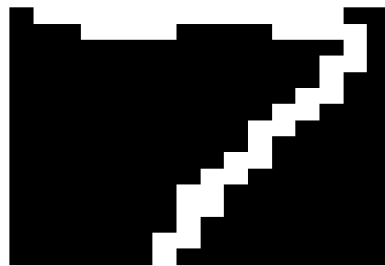
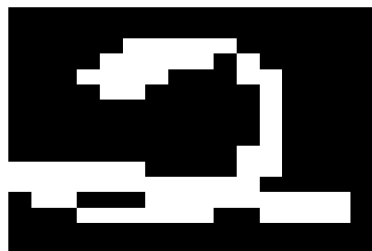
Exploratory Data Analysis

```
> sum(ziptrain27[,1] == 2);  
[1] 731  
> sum(ziptrain27[,1] == 7);  
[1] 645
```

As mentioned earlier, the dataset is formed by the response variable Y (labeled as V1 in R) and 256 covariates ranging from V2 to V257. Of the 1376 observations, there are 731 instances where Y = 2 and 645 where Y = 7.

```
rowindex = 5; ## You can try other "rowindex" values to see other rows  
> ziptrain27[rowindex,1];  
[1] 7  
Xval =  
t(matrix(data.matrix(ziptrain27[,-1])[rowindex,],byrow=TRUE,16,16)[16:1,]);  
image(Xval,col=gray(0:1),axes=FALSE) ## Also try "col=gray(0:32/32)"  
  
> rowindex = 16; ## You can try other "rowindex" values to see other rows  
> ziptrain27[rowindex,1];  
[1] 2  
> Xval =  
t(matrix(data.matrix(ziptrain27[,-1])[rowindex,],byrow=TRUE,16,16)[16:1,]);  
> image(Xval,col=gray(0:1),axes=FALSE) ## Also try "col=gray(0:32/32)"
```

This segment of code denotes the letter picture of the 5th row. In this case, two methods help establish the first image as the number 7: indexing and the image() function. For example, using row index = 16 would net us the second image and an indexing of the number 2.



Methods / Methodology

```
# training error
mod1 <- lm( V1 ~ . , data= ziptrain27);
pred1.train <- predict.lm(mod1, ziptrain27[,-1]);
ylpred.train <- 2 + 5*(pred1.train >= 4.5);
mean( ylpred.train != ziptrain27[,1]);
# testing error
pred1.test <- predict.lm(mod1, ziptest27[,-1]);
ylpred.test <- 2 + 5*(pred1.test >= 4.5);
mean(ylpred.test != ziptest27[,1]);
```

Linear regression is used to find a relationship between the Y value of 2 or 7, and the individual pixels. Using the midpoint method, any prediction that falls above or equals 4.5 would result in the classification of 7, and 2 if otherwise.

```
# training error knn
kk <- 1; # set to various levels of k. Start with 1 and end with 15.
xnew <- ziptrain27[,-1];
ypred2.train <- knn(ziptrain27[,-1], xnew, ziptrain27[,1], k=kk);
mean( ypred2.train != ziptrain27[,1])
# testing error knn
xnew2 <- ziptest27[,-1]; ## xnew2 is the X variables of the "testing" data
kk <- 1; ## below we use the training data "ziptrain27" to predict xnew2 via KNN
ypred2.test <- knn(ziptrain27[,-1], xnew2, ziptrain27[,1], k=kk);
mean( ypred2.test != ziptest27[,1]) ## Here "ziptest27[,1]" is the Y response of the "testing" data
```

KNN predicts the label of a data point by considering the labels of its nearest neighbors, with the number of neighbors specified by the variable, k. For instance, if k = 5, the model examines the five closest data points to predict. The KNN method is evaluated using different values of k.

Monte Carlo cross validation was employed to assess the models' reliability. In this approach, the training and testing datasets were merged and shuffled. The data was then randomly divided into a new training set (1376 samples) and a testing set (345 samples). This procedure was repeated 100 times, with the models being trained on each new training set and evaluated on the corresponding testing set. The error rates for each model were calculated and stored in a matrix, TEALL, and contain the testing error for the linear regression and various known models, which should also help to determine the ideal k value through the KNN method.

Results

Model	Training Error	Testing Error
LinReg	0.0007267442	0.0173913
KNN = 1	0	0.0173913
KNN = 3	0.01017442	0.01449275
KNN = 5	0.01235465	0.01449275
KNN = 7	0.01453488	0.0173913
KNN = 9	0.01598837	0.0173913
KNN = 11	0.01598837	0.0173913
KNN = 13	0.01744186	0.02028986
KNN = 15	0.01744186	0.02028986

The results compare the performance of linear regression (LinReg) and K-Nearest Neighbors (KNN) with different numbers of neighbors. Linear regression has a very low training error (0.0007) and a slightly higher testing error (0.0174), showing that it performs well on training and testing data.

For KNN, using $k=1$ perfectly fits the training data (0 training error) and has the same testing error as LinReg (0.0174). When $k=3$ or $k=5$, the testing error improves slightly to 0.0145, suggesting that these settings strike a good balance. However, as k increases, the testing error rises to 0.0203, meaning the model becomes too generalized.

A lower k value might be ideal, but comparing results based on training errors is difficult due to overfitting training data. However, based on the lowest value of testing error, we find that $k = 3$ or $k = 5$ to be ideal, as it is also smaller than the linear regression testing error.

Monte Carlo Cross Validation (refer to Appendix A)

Model	Mean Testing Error	Variance Testing Error
LinReg	0.011449	0.00002365601
KNN = 1	0.013420	0.00003032297
KNN = 3	0.014029	0.00002846698
KNN = 5	0.015768	0.00003858285
KNN = 7	0.016899	0.00004023516
KNN = 9	0.017449	0.00004310783
KNN = 11	0.018609	0.00004840338
KNN = 13	0.019913	0.00005408506
KNN = 15	0.020725	0.00005946887

Among all mean testing errors, linear regression has the lowest error. Linear regression is ideal for predicting correct classification with the test data. Similar to the previous discussion, the mean testing error increases as we increase the value of k because of underfitting.

Linear regression has the lowest variance testing error, which suggests that variability among different training and testing splits is very low. As for KNN, $k = 3$ would be the most balanced approach for achieving accuracy and balance, but $k = 1$ is also the most accurate based on mean testing error alone.

Appendix

Appendix A: MCCV Summary

```
> head(TEALL, 5)
```

	linearRegression	KNN1	KNN3	KNN5	KNN7
te	0.011594203	0.017391304	0.017391304	0.017391304	0.017391304
te	0.014492754	0.017391304	0.017391304	0.023188406	0.026086957
te	0.008695652	0.002898551	0.005797101	0.011594203	0.005797101
te	0.008695652	0.005797101	0.005797101	0.008695652	0.008695652
te	0.008695652	0.020289855	0.017391304	0.020289855	0.020289855

```
> summary(TEALL)
```

linearRegression		KNN1	KNN3
Min.	:0.002899	Min. :0.002899	Min. :0.002899
1st Qu.:	:0.008696	1st Qu.:0.008696	1st Qu.:0.008696
Median	:0.011594	Median :0.014493	Median :0.014493
Mean	:0.011449	Mean :0.013420	Mean :0.014029
3rd Qu.:	:0.014493	3rd Qu.:0.017391	3rd Qu.:0.017391
Max.	:0.023188	Max. :0.026087	Max. :0.026087

KNN5	KNN7	KNN9
Min. :0.002899	Min. :0.002899	Min. :0.002899
1st Qu.:0.011594	1st Qu.:0.011594	1st Qu.:0.014493
Median :0.014493	Median :0.017391	Median :0.017391
Mean :0.015768	Mean :0.016899	Mean :0.017449
3rd Qu.:0.020290	3rd Qu.:0.020290	3rd Qu.:0.023188
Max. :0.034783	Max. :0.034783	Max. :0.034783

KNN11	KNN13	KNN15
Min. :0.005797	Min. :0.005797	Min. :0.005797
1st Qu.:0.014493	1st Qu.:0.014493	1st Qu.:0.014493
Median :0.017391	Median :0.020290	Median :0.020290
Mean :0.018609	Mean :0.019913	Mean :0.020725
3rd Qu.:0.023188	3rd Qu.:0.023188	3rd Qu.:0.023188
Max. :0.040580	Max. :0.040580	Max. :0.043478